```
#Logistic Regression
import seaborn as sns
import pandas as pd
df=sns.load_dataset("titanic")
df
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	True
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	True
888	0	3	female	NaN	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	False
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	True
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	True

891 rows × 15 columns

Next steps: Generate code with df New interactive sheet

```
x=df.drop(['class', 'who','adult_male','deck','embark_town', 'alive', 'alone', 'survived' ], axis=1)
x = pd.get_dummies(x, columns=['sex', 'embarked'], drop_first=True)
x = x.fillna(x.mean(numeric_only=True))
```

```
y=df["survived"]
```

```
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    x, y,
    test_size=0.2,
    random_state=42
)
```

```
from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()
model.fit(x_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

```
df["age"]=df["age"].fillna(df["age"].mean())
df.drop(["deck", "alive"], axis=1, inplace=True)
df["embark_town"]=df["embark_town"].fillna(df["embark_town"].mode()[0])
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	0
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
embark_town	0
alone	0

dtype: int64

```
df=pd.get_dummies(df, drop_first=True)

x=df.drop("survived", axis=1)
y=df["survived"]
```

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
```

```
from sklearn.linear_model import LogisticRegression
log_model = LogisticRegression(max_iter=1000)
```

```
log_model.fit(x_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred = log_model.predict(x_test)

y_prob = log_model.predict_proba(x_test)
```

```
from sklearn.metrics import accuracy_score
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.8156424581005587
```

```
from sklearn.metrics import confusion_matrix
conf_matrix = confusion_matrix(y_test, y_pred)
print(conf_matrix)
```

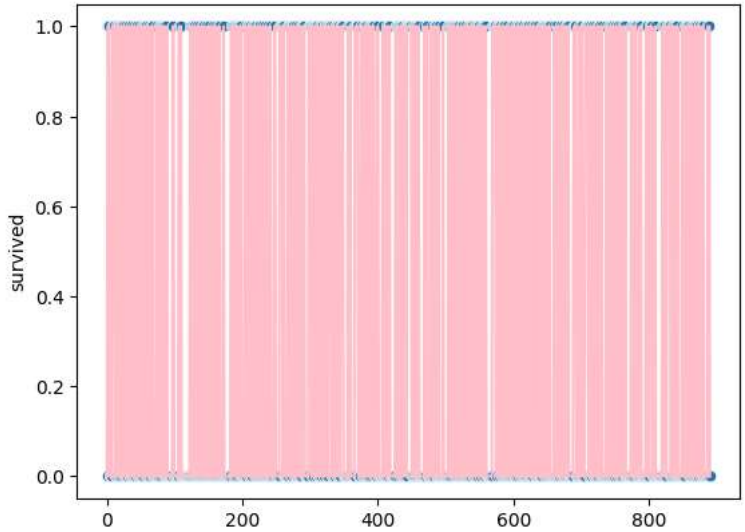
```
[[91 14]
 [19 55]]
```

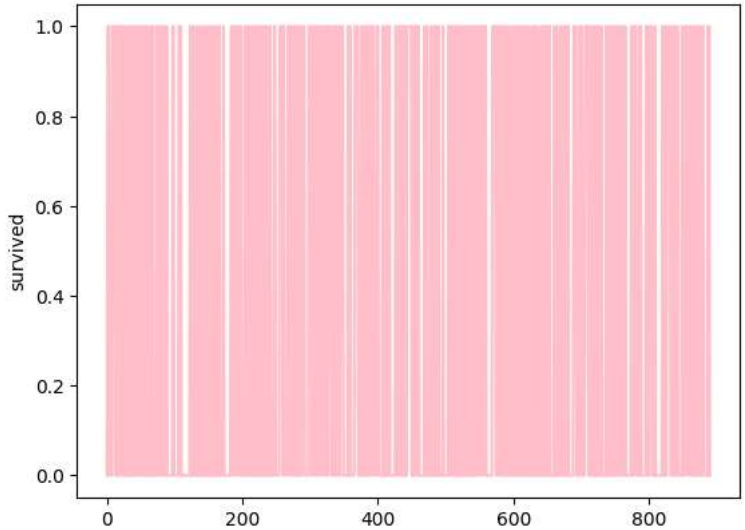
```
from sklearn.metrics import confusion_matrix
conf_matrix = confusion_matrix(y_test, y_pred)
print(conf_matrix)
```

```
[[91 14]
 [19 55]]
```

```
sns.scatterplot(y)
sns.lineplot(y, color='pink')
```

```
<Axes: ylabel='survived'>
```




```
sns.lineplot(y, color='pink')
```

```
<Axes: ylabel='survived'>
```